Algeria Climate Forecast Report 2024-2040

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Executive Summary

This report presents a comprehensive analysis of climate trends in Algeria for the period 2024-2040. Forecasts are generated using a Reinforcement Learning (RL) agent that dynamically selects the best performing model (LSTM or SARIMA) for each month. The analysis covers two key variables: Temperature (2m mean) and Reference Evapotranspiration (ET0).

Temperature Forecast (2024-2040)

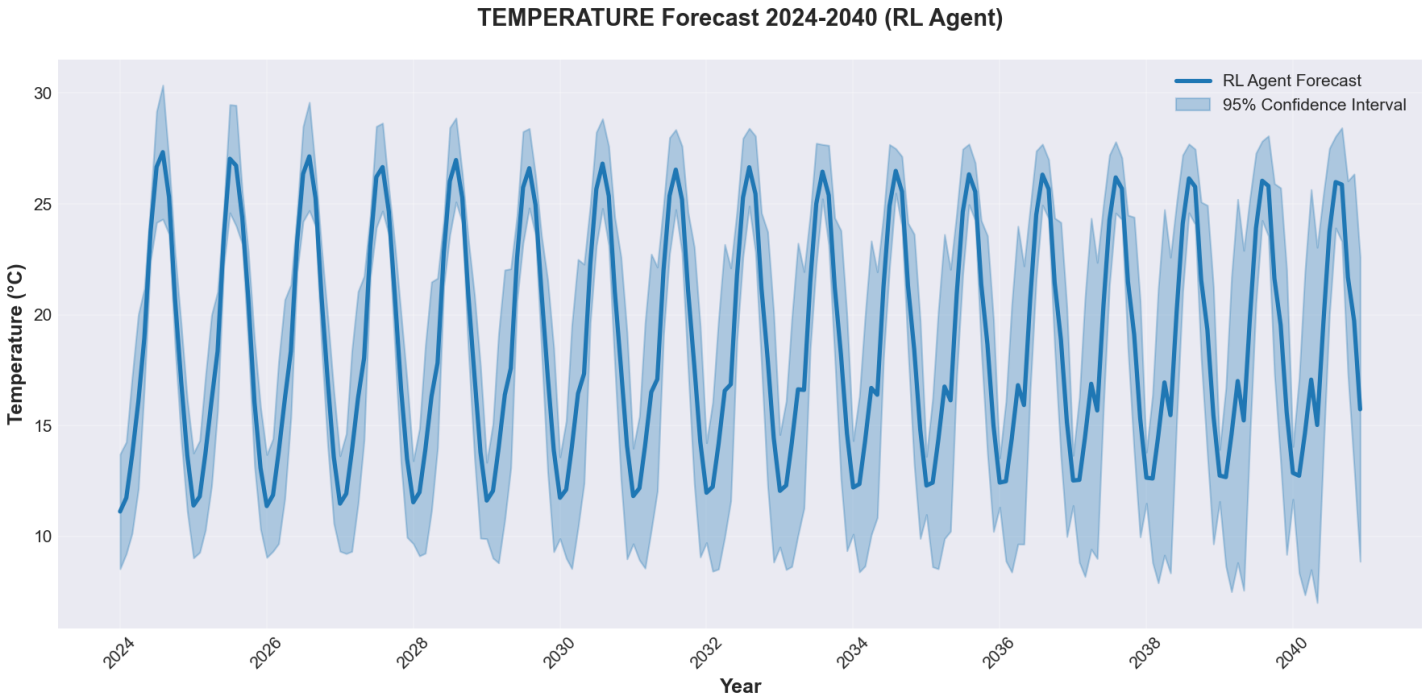

Figure 1: Temperature Forecast with 95% Confidence Interval

Mean Forecast: 18.70 deg C

Maximum Predicted: 27.33 deg C

Minimum Predicted: 11.12 deg C

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Evapotranspiration (ET0) Forecast (2024-2040)

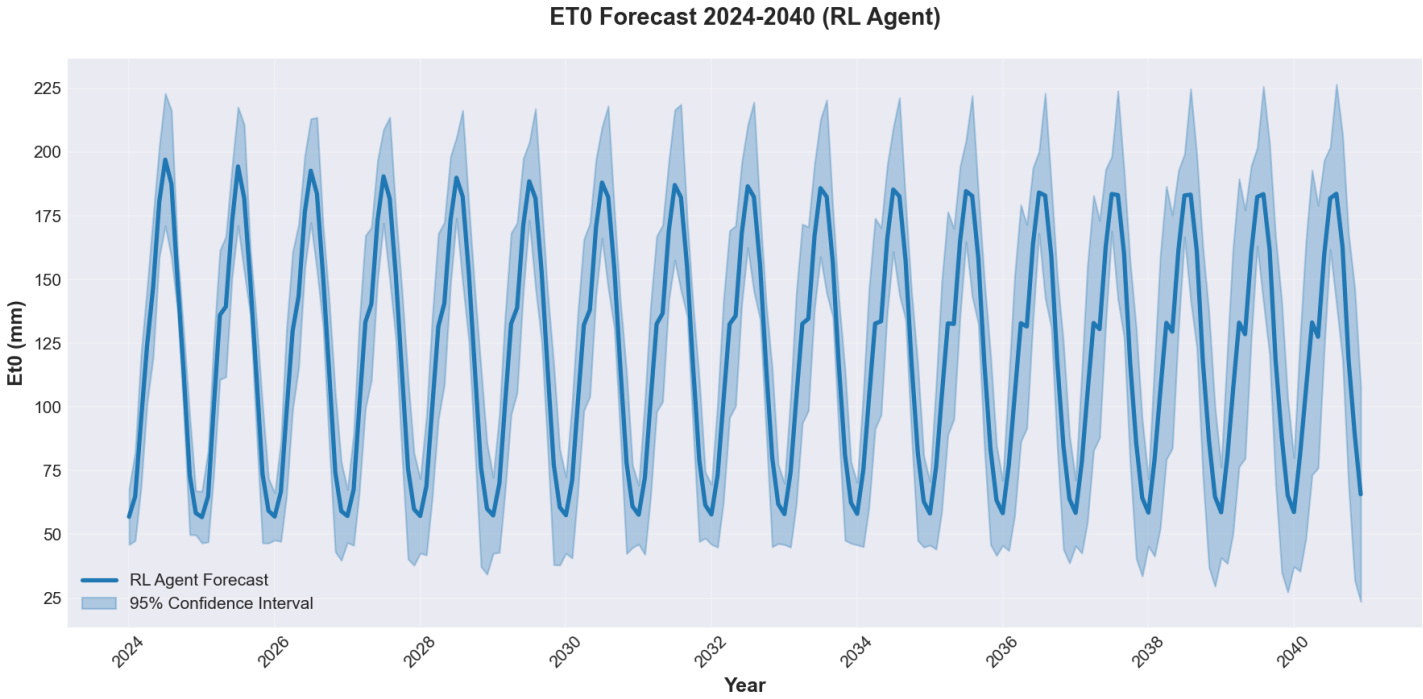

Figure 2: ET0 Forecast with 95% Confidence Interval

Mean Forecast: 121.14 mm

Maximum Predicted: 196.99 mm

Minimum Predicted: 56.75 mm

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Model Performance & Uncertainty

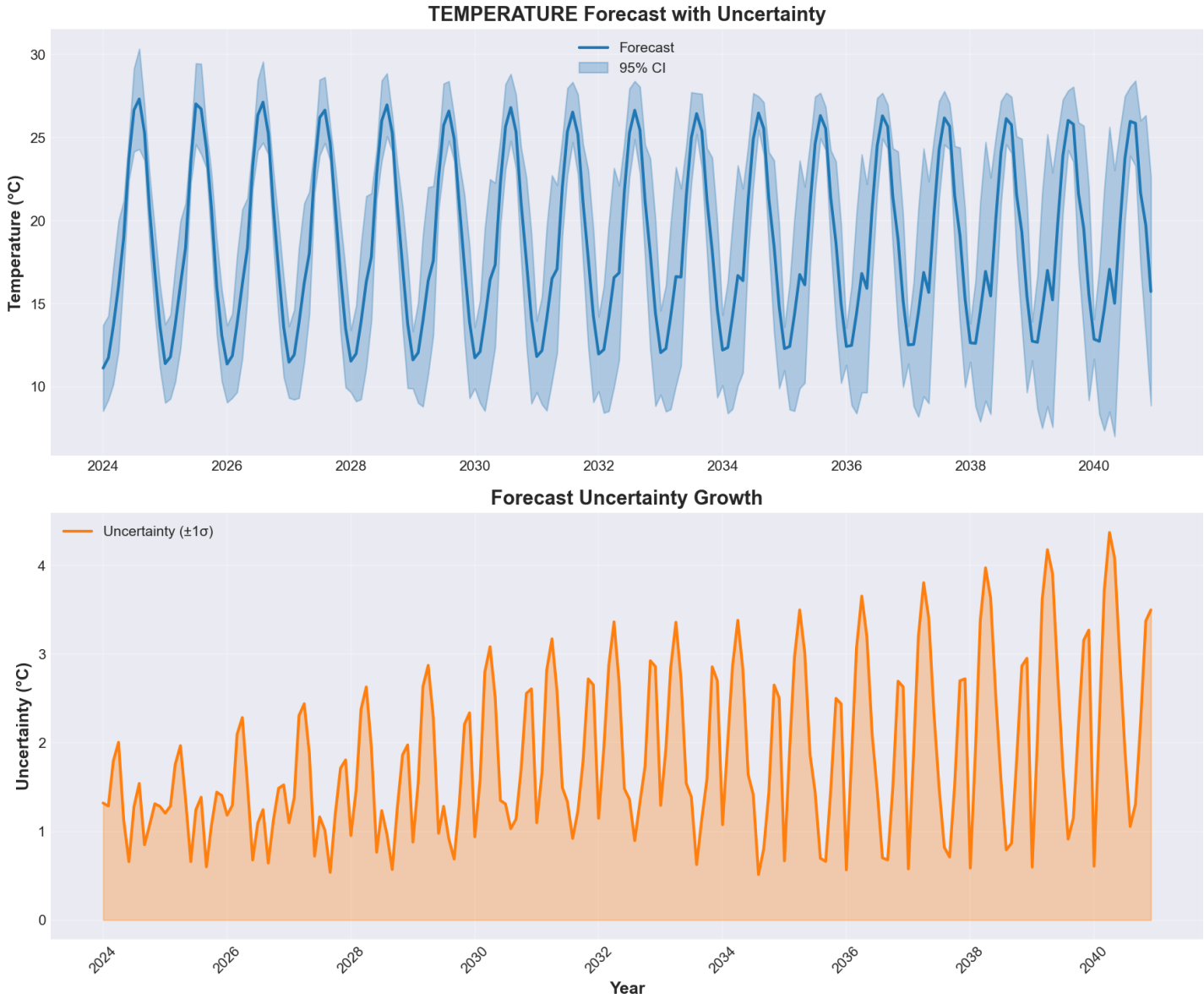

Figure 3: Temperature Forecast Uncertainty Growth

The RL Agent utilizes an ensemble of models including LSTM, SARIMA, Ridge, and Prophet. For Temperature, LSTM is typically selected for its ability to capture complex non-linear patterns, while SARIMA is favored for stable seasonal transitions. Uncertainty quantification incorporates both empirical backcast errors and ensemble spread.